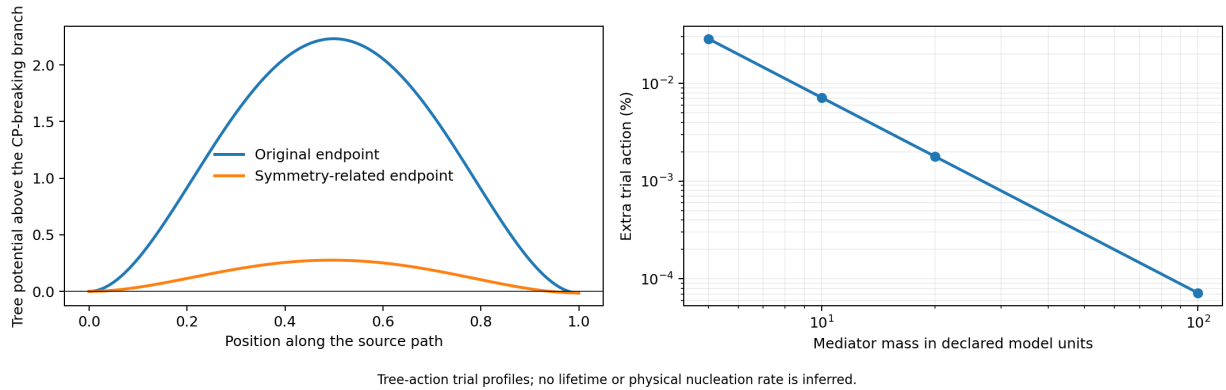


Polynomial certification in flavor physics: decay, EFT control and CP constraints

The next advance closes the matrix interface needed to apply the polynomial machinery to coupled flavor physics. The new module `python/perfectpower/fluxor_matrix_response.py` computes exact polynomial determinants by fraction-free Bareiss elimination, compiles supplied relaxed responses using a bordered determinant, and establishes positive matrix and operator-norm bounds through principal minors. It consumes the other flavor session's finite Gaussian matching receipt and changes none of that session's loop, pole or mechanism calculations. The driver is `python/develop_fluxor_matrix_response.py` and its result is `receipts/fluxor_cosmology/matrix_response.json`. Run `PYTHONPATH=python python python/develop_fluxor_matrix_response.py` to regenerate it; add `--verify` to replay the persisted packet without positivity discovery. The same focused flavor test command given below now runs forty-six tests, including fifteen new matrix tests.

Published fluxor action: trial decay paths and mediator kinetic cost



Tree-action trial profiles; no lifetime or physical nucleation rate is inferred.

The first application is to both published 21-coordinate derivative metrics. Let E be the symmetrized rational realization of $K(0)-I$ at reference mediator mass 10. For the finite 27-mediator completion, the exact matrix norm is strictly below $567/500000$, which is 0.001134. For the universal 55-mediator completion it is strictly below $321/250000$, which is 0.001284. These are bounds in every scalar direction, established by positive principal minors of both $b \ I+E$ and $b \ I-E$. They do not merely compare a few sampled eigenvectors. The retained matrices consist of exact binary fractions of the published floats, with each off-diagonal pair averaged rationally. This explicit symmetrization is the only change from the published matrix arrays. The result is exact for those rational matrices, not a reconstruction of the exact algebraic finite-group projectors.

The up/down source coupling block is bounded independently. If B is the ten-by-ten cross-sector block, positivity of b squared $I-B$ transpose B proves its largest singular value is below b . The finite completion has bound $391/1000000$, or 0.000391; the universal completion has bound $83/200000$, or 0.000415. Thus the same Gaussian completion that correlates source potentials also controls a complete scalar cross-sector normalization map. Forty-two principal minors per full metric and ten per cross-sector block are checked, for 104 exact positive principal-minor values across the two published completions. A scalar metric is not a quark kinetic matrix: these bounds are not fed into a CKM angle without the separate matched fermion calculation.

The envelopes transport continuously in mediator mass on the declared fixed-EFT Gaussian valley. The current Jacobian scales with M when its exchange quartics are held fixed, so the metric correction scales as $1/M$ squared. For every $M>0$ the retained matrix norm is therefore below $(10/M)$ squared times the corresponding reference bound, and its cross-sector norm has the same scaling. Write $\epsilon=(10/M)$ squared b . If $\epsilon<1$, every eigenvalue of the retained metric K_M lies between $1-\epsilon$ and $1+\epsilon$. In particular it is positive for M squared greater than 0.1134 in the finite case and 0.1284 in the universal case. These are conservative sufficient conditions for the rounded matrix, not physical lower mediator-mass limits. The exact Gaussian construction already has a positive induced metric from its Gram form; it is unnecessary to identify tiny roundoff directions of a retained array with physical instabilities.

For any separately supplied positive source Hessian H , that normalization envelope implies $\lambda_i(H)/(1+\epsilon) \leq \lambda_i(H, K_M) \leq \lambda_i(H)/(1-\epsilon)$, where the middle quantity is the ordered generalized eigenvalue in H

$v = \lambda K_M v$. The proof is the Rayleigh quotient bound followed by the min-max principle. The present matrix certificate does not establish the source Hessian's eigenvalues or replace the other session's exact finite-momentum pole equation. It gives a conditional all-direction normalization bound, useful before solving that equation or checking a derivative expansion. Quantum corrections, heavy thresholds and the unclassified global flavor vacuum remain separate questions.

The central reusable calculation is the relaxed response $R(x) = d(x) - g(x) \text{ transpose } H(x) \text{ inverse } o(x)$. Here g is a supplied observable gradient, o is the gradient of a supplied interaction, and d is its direct matched response. The entries can be rational polynomials in one parameter. Let $D = \det H$ and C be the determinant of the bordered matrix with rows (H, o) and $(g \text{ transpose}, 0)$. The block determinant identity gives $C = -D g \text{ transpose } H \text{ inverse } o$ wherever D is nonzero, so $R = (dD + C)/D$. This needs two determinant computations rather than a numerical inverse or all individual cofactor matrices. Rational functions of the parameter are avoided inside the polynomial elimination by exact division by the preceding pivot. Exact row pivoting handles vanishing polynomial pivots, while a degree budget and dimension budget make unsupported inputs explicit.

An important branch condition survives compilation. The denominator remains $\det H$ even if the numerator is identically zero or shares factors with it. For $H = ((1, x), (x, 1))$, $g = (1, x)$, $o = (1, 0)$ and direct response one, the compiled numerator is exactly zero while the denominator remains $1 - x^2$. Cancellation has removed this particular response; it has not made the Hessian invertible at $x = 1$. The implementation refuses to certify that matrix as stable on the whole halfline. In flavor applications, a reduced Hessian must first remove gauge directions, and a supplied stationary background must already satisfy the physical matching hypotheses. An inverse-Hessian calculation cannot supply either fact by itself.

Matrix stability over a continuous interval is implemented as well. A symmetric polynomial H is positive definite if each leading principal minor is strictly positive. This follows from its pivoted quadratic-form factorization: the successive diagonal factors are positive ratios of those minors. To prove a scalar polynomial f positive on a closed rational interval $[a, b]$, transform it to $(1+t)$ to the degree times $f((a+b t)/(1+t))$ on $t \geq 0$, and check $f(b) > 0$ separately. The mapping covers $[a, b]$ and the separate endpoint closes the interval. The existing exact halfline Sturm operation then applies. This adds finite-interval response tests without sampling near a pole or extending a local claim beyond its stated branch. Positive-semidefinite boundary points and singular gauge-reduced limits are deliberately outside this strict interface.

The operation `protected_response` combines those matrix stability conditions with a uniform absolute response bound. If $\beta^2 D^2 - N^2$ is strictly positive throughout the same domain, then $|N/D| < \beta$ there; positivity also excludes denominator zeros. A complete packet therefore contains the supplied Hessian, gradients and direct term, its bordered determinant, the exact response numerator and unreduced denominator, every positive principal minor and the response-budget polynomial. This is ready for actual matched strong-CP or CKM gradients from the other session. It makes no such gradient up: the release's applied metric bounds are physical scalar data, while its response demonstrations are explicitly identified propagator and algebraic examples.

The first exact response example reuses the other session's rational Gaussian fixture: $H = ((3, 1/3), (1/3, 2))$, three current rows $(1, 2)$, $(2, -1)$, $(1, 1)$, and squared mediator masses 11, 13, 17. Its local two-derivative Euclidean operator is $F(t) = H + t K(0)$, $t \geq 0$. Taking $g = (1, 0)$, $o = (0, 1)$, $d = 0$ gives $N = 1/3 + (48193/5909761)t$ and $D = 53/9 + (92062063/17729283)t + (6378192/5909761)t^2$. The relaxed response equals positive $3/53$ at zero momentum; the propagator's cross entry itself has the opposite sign. Exact matrix and budget polynomials prove $F(t)$ positive and $|R(t)| < 3/50$ for every nonnegative Euclidean squared momentum. This is an exact statement about the local two-derivative fixture, not a substitution of a rational kernel for the physical B_0 loop function or the finite nonlocal propagator.

The second response demonstration uses $H = ((1, t), (t, 1))$, $g = o = (1, 0)$ and $d = 0$ on $[0, 1/2]$. Its response is $-1/(1-t^2)$; the stable-interval certificate proves its magnitude below $27/20$ throughout that closed interval. A budget of one is rejected, as is extending stability through the pole at $t = 1$. These examples establish the matrix-to-response plumbing independently of an inferred flavor prediction. In particular, first-order phase protection from a Hermitian source remains the other session's theorem; the later determinant-phase and regenerated-portal obstructions in its finite matching chapter remain untouched.

The fifteen new tests compare polynomial Bareiss determinants with independent permutation expansions, include row pivots and identically singular matrices, check explicit inverse-Hessian values at rational points, retain a canceled singular denominator, and test continuous matrix stability, both interval endpoints, interior zero touches, spectral envelopes and cross-sector Gram bounds. They also check the complete applied 21-field packet against its pinned source receipt. All forty-six focused tests pass. Regeneration and publication validation are recorded separately, with no claim of a full repository regression or a new Lean theorem. This matrix layer now makes the highest-value handoff concrete: the matching session can supply reduced Hessian and response polynomials; the polynomial layer can turn those into continuous stability and flavor-response budget statements on a declared branch.

The polynomial machinery has its highest immediate impact as a certification layer between matched flavor mechanisms and their parameter scans. An optimizer proposes a point, while exact signed polynomial remainders can establish a statement over an entire continuous parameter range. This separates numerical discovery from an exclusion or bound that can be replayed. The implemented bridge uses the repository's existing rational polynomial and Sturm core, without changing the other session's source tensors, matching, scalar poles, running or vacuum discovery. Its first physical application closes the previously open optimization question for every retained compact decay-profile family and supplies a uniform bound on the mediator kinetic correction.

The new interface is `python/perfectpower/flavor_polynomial_bounds.py`. It provides `positive_halfline`, `rational_lower_bound` and `rational_absolute_budget`, together with verifiers that check supplied signed remainders rather than repeating root discovery. The driver is `python/develop_flavor_polynomial_bounds.py`; the receipt is `receipts/flavor_cosmology/polynomial_bounds.json`. Run `PYTHONPATH=python python python/develop_flavor_polynomial_bounds.py` to rebuild it; add `--verify` for discovery-free replay of the persisted receipt. Run `PYTHONPATH=python python -m unittest discover -s python/tests -p 'test_flavor*.py'` for the focused checks. The report verifier also checks the source receipt's byte hash, binds every rational numerator and denominator to its supplied path polynomials, and checks the exact mediator factors and ratio intervals. The new tests include corrupted remainders, truncated chains, a wrong source hash, a changed mediator mass and altered polynomial coefficients.

For a real polynomial f with rational coefficients and $f(0) > 0$, construct the Sturm chain starting from f and its derivative, then successive negative Euclidean remainders. Each member is multiplied by a positive rational factor to obtain primitive integer coefficients. This preserves its signs. Sign variations at zero use each member's constant coefficient; at positive infinity they use its leading coefficient. Equal variation counts prove that f has no positive real root. Together with $f(0) > 0$ and continuity, this proves $f(x) > 0$ on the entire closed halfline. The replay checks the initial polynomial, derivative, every signed remainder, the terminal zero remainder and the two variation counts. Repeated roots are supported by the terminal gcd; a polynomial touching zero at a positive point is intentionally rejected. Positive constants are handled separately. Failed certification is inconclusive, including cases where an inequality is true only on a restricted interval. There is no automatic promotion to a general semialgebraic decision procedure or a Lean proof.

For a supplied rational observable $F(x) = N(x)/D(x)$, positivity of $N - bD$ on the entire halfline proves $F > b$ at every $x \geq 0$ with $D(x) > 0$. Importantly, D need not be positive everywhere. A feasible rational witness supplies a finite upper bound on the infimum. This is exactly the structure of the scale-reduced trial action: $N = T$ squared, $D = -2U$, and $F = S_{\max}/\pi$ squared. The previous numerical shape is used only as a witness and a proposal for b . The proof covers every positive-denominator component, arbitrary large L and all narrow unsampled regions. No numerical stationary-root reconstruction, finite search interval or rounded root matching enters the result.

The driver accepts a lower bound equal to the exact witness value times $(1 - 10^{-10})$. All five retained source-path certificates pass, as do the four lifted paths at mediator masses 5, 10, 20 and 100. The best retained source family's infimum lies between 33,396,009.266787387 and 33,396,009.270126987 in dimensionless action units. These displayed decimal endpoints are approximate renderings of exact rational bounds multiplied by π squared; the rational endpoints divided by π squared in the receipt are authoritative. The ratio of the original endpoint family's infimum to that of the best retained family's infimum lies between 757.735824000410 and 757.735824151957. The three almost identical best endpoint images are not assigned a unique ordering when their certified intervals overlap. This result certifies a family-level variational improvement, not completeness of the symmetry-path search or the true field-space bounce.

There is also an all-shape, all-mass control statement. Let $\delta(L)$ be the exact kinetic increase at reference mass 10. The source kinetic polynomial T and δ are both proved strictly positive for every $L \geq 0$. At fixed effective quartic

couplings the same Gaussian valley has $T_M = T + (10/M)^2 \delta$. Thus the lifted scale-reduced action exceeds the source action at every admissible shape, increases as the mediator mass decreases, and its infimum is nondecreasing. The mass-10 optimum-to-optimum ratio is bounded between 1.0000715282043315 and 1.0000715284043460. At masses 5, 20 and 100 the corresponding intervals are 1.0002861284656723 to 1.0002861286657296, 1.0000178817362650 to 1.0000178819362686, and 1.0000007151703810 to 1.0000007153703812. All intervals are stored as exact fractions as well as display decimals.

The stronger uniform statement uses `rational_absolute_budget`. Positivity of $b^2 D^2 - N^2$ proves $|N/D| < b$ and also excludes every denominator zero on the halfline: at a zero of D the residual could not be positive. Applied to $N = \delta$, $D = T$ and $b = 361/10000000$, it proves $\delta/T < 0.0000361$ for every shape. Consequently $1 < S_M(L)/S_{\text{source}}(L) < [1 + (10/M)^2 \cdot 0.0000361]^2$ for every $M > 0$ and every admissible L . At mass 10 the upper ratio is exactly 1.00007220130321, or an increase below 0.007220130321 percent. The same non-strict envelope bounds the ratio of the two positive infima. This controls the kinetic error uniformly on the supplied valley. It does not control transverse paths, running, loop backreaction or higher operators; in particular it does not repair the previously identified uncontrolled quantum displacement of this benchmark.

The next high-impact uses are precise interfaces for the active matching session. A relaxed first-order flavor response has a supplied direct term minus gradient Q transpose H inverse gradient O . On a one-parameter rational coefficient slice, an exact reduced Hessian and exact gradients allow adjugate/determinant elimination to express that response as N/D . The absolute-budget certificate can then bound a strong-CP response or a matched CKM response over the whole slice, and rule out poles at the same time. This requires the other session to supply the actual matched observable, canonical gauge-reduced Hessian and a justified expansion domain. A bound on a linearized response remains a bound on that response; it is not an all-orders strong-CP theorem, a symmetry explanation for tuned coefficients or a predicted CKM phase. The module already exposes the rational budget operation, but this release deliberately assigns no actual strong-CP derivative to a toy numerator.

A second use is vacuum and pole exclusion along specified coefficient slices. With an exact symmetric reduced Hessian, positivity certificates for all leading principal minors give a sufficient positive-definiteness statement by Sylvester's criterion. Supplied characteristic polynomials and Sturm variation counts can likewise exclude real eigenvalue crossings in a specified mass-squared interval. Multiple parameters, algebraic-number source tensors, gauge null directions and singular matching denominators need their own adapters; the present rational halfline implementation does not silently resolve them. These are integration targets rather than new scalar-pole or vacuum-discovery calculations, which remain owned by the other flavor session.

The release produces twelve independently replayable positivity certificates, thirteen new focused tests and eighteen existing decay tests, all executed successfully. It additionally replays the full report against the original tree-decay receipt and checks deliberate source and certificate corruption. No new Lean theorem or full repository regression is claimed. The exact statements apply to the retained binary-rational tensor realization and supplied paths, not to a reconstructed exact finite-group action or formally stationary endpoints. A lower bound on the optimized trial family is not a lower bound on the least physical bounce action and cannot establish longevity. Publication is directly on the consolidated working branch, preserving the incoming Gaussian-pole and elliptic-division advances.

The preceding decay calculation and conditional CP-wall interfaces follow with their original assumptions. Its numerical one-parameter optimization limitation is now replaced by the family-level certificates above; its endpoint, global-vacuum and quantum limitations remain.

The preceding release turns the conditional wall interface into a calculation on the published flavor action. It consumes the lower CP-conserving competitor already found by the other flavor session, rather than searching for new stationary points or changing its coefficients. At the pinned main-branch commit `22e4edf06dbdeea0f9245dbb06669ed57b7906fc`, the nominated CP-breaking branch has energy -4.196243686677212 , while the retained stable CP-conserving competitor has energy -4.210439239580322 . The gap is 0.01419555290310992 . Both are locally stationary to approximately 10^{-15} in the declared numerical action, and their reported smallest physical Hessian eigenvalues are positive. The absolute global minimum remains unclassified. This benchmark therefore calls for a metastable-decay calculation, not a claim of stable walls between global CP-breaking vacua.

The new reusable module is `python/perfectpower/flavor_decay.py`, with the driver `python/develop_flavor_cosmology.py` and eighteen focused tests in `python/tests/test_flavor_decay.py`. The scientific receipt is `receipts/flavor_cosmology/tree_decay.json`. It retains exact rational coefficients for the radial trial profiles, the input-file hashes, every chosen endpoint and the full 48-coordinate mediator-lifted path. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python python/develop_flavor_cosmology.py` from the repository root. No source couplings are retuned, no stationary points are re-minimized, and no quark matching is repeated. The earlier 104-check Node calculation remains an independent conditional interface.

The numerical source tensors, coefficients and endpoint coordinates are converted to exact binary fractions. Five exact evaluations along a source line reconstruct its quartic potential by rational Newton interpolation. This arithmetic is exact for the binary-rational realization of those arrays. It does not turn floating-point finite-group projectors into exact algebraic-number tensors, or approximate stationary coordinates into proven exact minima. Thirty-three independent ordinary floating-point evaluations per candidate check the reconstructed polynomial against the existing full potential. This distinction matters: the radial action integrals below are exact, but the physical identification with a certified bounce remains conditional on the false-vacuum and reduced-variational hypotheses.

Let $\gamma(q)$ be a polynomial path in canonically normalized real fields, with $\gamma(0)$ the supplied false vacuum and $\gamma(1)$ a lower-energy configuration. Let $W(q)=V(\gamma(q))-V(\gamma(0))$. A compact $O(4)$ trial profile has $q=1$ inside a core, a smooth shell, and $q=0$ outside. Put $u=r/R$. The core is $0 \leq u \leq L$; in the shell write $t=u-L$ between zero and one and set $q(t)=1-3t^2+2t^3$. This taper has zero derivative at both joins, so the profile is continuously differentiable. Outside $u=L+1$ the field equals the false background exactly. No exponential tail cutoff or divergent outside energy is introduced.

The complete action of this profile is

$$S_4(R,L)=2\pi^2 [T(L) R^2 + U(L) R^4],$$

where $T(L)=\text{one half times the integral from zero to one of } (L+t)^3 \text{ times the squared norm of } d\gamma(q(t))/dt$. The potential term is $U(L)=W(1)L^{4/4}$ plus the integral from zero to one of $(L+t)^3 W(q(t))$. Both are rational polynomials in L whenever the supplied field-path and potential coefficients are rational. T has degree at most three and U has degree at most four. The core-volume contribution ensures $U<0$ for sufficiently large L whenever $W(1)<0$. This makes a negative-action escape configuration available without pretending the Euclidean bounce is a global action minimum.

For a retained shape with $T>0$ and $U<0$, the action as a function of size starts at zero, reaches a positive maximum and then becomes negative. The maximum lies at $R_{\text{crit}}^2=-T/(2U)$, with $S_{\text{max}}/\pi^2=-T^2/(2U)$. Its derivative under radial scaling vanishes there and its scaling second derivative is negative, as expected for the escape direction. The code evaluates these quantities as exact fractions. It uses a numerical one-parameter search only to choose a rational L and then recomputes the retained action exactly. It does not certify global optimality even within the shape family.

The appropriate variational interpretation is an upper estimate for the least decay barrier, under the usual reduced-problem or mountain-pass existence assumptions and an exact metastable exterior. Every such trial profile produces a route from the false configuration to negative action; the least separating saddle cannot have a larger minimax level than that route. This is not a lower bound on the bounce action. Consequently a large trial action cannot establish a long lifetime. The familiar scale-reduced formulation originates in Coleman, Glaser and Martin, Action Minima among Solutions to a Class of Euclidean Scalar Field Equations, Communications in Mathematical Physics 58 (1978) 211-221, DOI 10.1007/BF01609421. The multifield symmetry and reduced-problem existence assumptions are treated by Blum, Honda, Sato, Takimoto and Tobioka, $O(N)$ Invariance of the Multi-Field Bounce, arXiv:1611.04570, <https://arxiv.org/abs/1611.04570>. The present radial-profile construction and its integration are an adapter to this established framework.

A significant practical reduction comes from correctly using the internal symmetry. The driver evaluates the 1080 existing family matrices on the already-published lower endpoint, together with both sector-sign transformations. This supplies 4320 symmetry-related endpoint images; it does not produce 4320 independently minimized vacua. The source coefficients remain identical. Four nearest images and the original endpoint are retained as candidate straight paths. A central group image can repeat the same real-adjoint endpoint, so the retained list is not claimed to classify distinct physical vacua. A farther endpoint might have a better potential barrier; this finite nearest-image selection is deliberately

a bounded path search, not an exhaustive optimization.

With the original endpoint, the retained compact-profile trial action is approximately 2.5305352605 times 10^{10} . The best retained symmetry-related endpoint yields approximately 3.3396009270 times 10^7 , a reduction by a factor of 757.7358241. Its squared source distance is 3.210476017821027; the original endpoint's squared distance is 11.166145333203302. Its core-to-shell ratio L is 45.94751799. These are dimensionless trial actions for the specified tree potential. Their numerical size is not a cosmological lifetime, a measured decay rate, or a statement that this action is near the actual least bounce. The result shows why symmetry-aware field-path selection materially affects even a conservative decay calculation.

The mediator completion introduces a further geometric issue. The positive Gaussian square allows the scalar potential to match the source potential exactly at $S=-J(x)/M^2$, but this equality does not erase the mediator kinetic energy when x varies across a bubble. Following that trough gives the induced field-space metric

$$ds^2 = dx^T dx + dJ^T dJ/M^4.$$

The driver constructs all 27 mediator coordinates as exact quadratic polynomials along the source line, retaining the 21 source/Higgs coordinates. The complete trial bubble therefore uses 48 canonical real fields. Its potential along the path is unchanged, while its kinetic integral increases by one half the radial integral of the squared mediator derivatives. The positive-square contribution is evaluated independently at seventeen path points; it is negligible at floating-point reconstruction precision. Potential matching and kinetic matching are thus kept separate.

At any fixed path and retained shape, the completion changes the candidate action by the exact factor $[1+\Delta T(L)/T_{\text{source}}(L)]^2$. The extra kinetic integral is nonnegative, so this factor cannot decrease the action of that particular trial profile. This is not a theorem that the true UV least bounce is greater than the EFT least bounce: UV paths can leave the trough, and the least path has not been solved. The statement concerns the explicitly retained trough-following profiles. Treating Gaussian potential matching alone as sufficient for tunneling would omit this positive geometric cost.

The mass scan keeps the EFT coefficients fixed and scales the current couplings in the existing Gaussian realization proportionally to M . Then $S=-J/M^2$ scales as $1/M$, and the induced kinetic correction scales as $1/M^2$. For masses 5,10,20,100 in the stated model units, the separately shape-optimized trough-profile action ratios to the best source candidate are approximately 1.0002861286, 1.0000715283, 1.0000178818, and 1.0000007153. At $M=10$ the lifted action is approximately 3.3398398030 times 10^7 , an increase of 0.00715283 percent. This scan demonstrates decoupling within the same completed tree construction. It does not vary the mediator mass while incorrectly freezing its current normalization, and it includes no mediator-loop Coleman–Weinberg correction.

The CP interpretation also becomes sharper. Direct evaluation of the supplied generalized-CP partner reproduces the false-branch energy, and CP covariance is checked on the false point, chosen lower point and their midpoint. CP-even matching or subsequent CP-even thermal evolution cannot select a weak-CP sign merely by splitting this conjugate pair. If the CP-conserving lower basin ultimately occupies the universe, it removes the physical weak-CP order parameter instead of predicting its observed sign. Whether this basin is populated requires nucleation dynamics, initial conditions and an adequately controlled action; the local CP-breaking benchmark alone supplies none of those.

The upstream quantum calculation is a substantial limit on physical interpretation. At its fixed $\mu=1$ inputs, the linearized one-loop vacuum displacement is 15.51581387 times the tree-background norm. This signals an uncontrolled small-displacement expansion. A tree trial action cannot be presented as the quantum-corrected tunneling exponent for that benchmark. The current result deliberately imports and displays that warning alongside its own action rather than fitting counterterms or changing couplings to conceal it. A controlled quantum action is an input the mechanism session must eventually supply before physical decay claims become meaningful.

The upstream leading high-temperature minimal-action Hessian is positive, supporting asymptotic restoration within its declared expansion. That does not provide a transition temperature or an $O(3)$ nucleation action. The new calculation is zero-temperature, flat-space $O(4)$, with gravity omitted. A finite-temperature extension requires the resummed effective potential, the thermally active mediator content, possible friction and the maximum post-inflation temperature. Inferring thermal bubble rates by dividing the present $O(4)$ action by temperature would be incorrect. Likewise, reinstating a

stable-wall calculation requires a candidate whose CP-breaking vacua actually lie on the global branch.

The eighteen Python tests independently integrate the compact shell with numerical quadrature, check rational interpolation, endpoint smoothness, action-scaling stationarity and the negative-action escape, reject invalid cores and exterior energies, prove scale cancellation on exact fixtures, and check the increased kinetic cost of extra fields. Receipt tests verify the published energy ordering, exact retained action replay, full-potential sample agreement, the 48-field lift, mediator-current matching, monotone decoupling in the retained scan, and CP degeneracy. The scientific receipt replays byte for byte in a fresh process under the specified single-thread numerical environment. These are focused mathematical and action checks; no full repository regression or new Lean theorem is claimed for this push.

The continued coordination boundary is explicit. The other flavor session owns exact tensor closure, constrained mechanisms, running, quantum corrections, physical matching and vacuum discovery. This release consumes its c9010fb conclusions and changes only the cosmology module, driver, focused tests, cosmology receipt, plotting/rendering scripts and this monograph. It commits directly to the main working branch after checking the latest head. The arithmetic and Lean streams remain intact. Future cosmology calculations can reuse the same radial compiler on a new action without borrowing protection, hierarchy or vacuum claims from incompatible branches.

The advance is a reusable decay-path compiler tied to the repository's actual published action, together with its mediator-induced kinetic geometry and exact radial witnesses. It sharpens what has to be measured next: a controlled action and least-path calculation, a dimensionful normalization and prefactor, and a viable cosmological occupation history. The conditional CP-wall and constrained-bias results from the preceding release remain useful for future globally CP-breaking candidates; they are retained below with their original hypotheses.

This advance consumes the published flavor conclusions without changing their mechanisms, matching calculations, operator algebra, scalar vacua, or cumulative flavor monograph. It adds exact rational tools for a conditional cosmological question: if the chosen flavor branch really supplies degenerate physical CP-breaking global vacua, can a proposed cosmological history avoid their walls, and can a permitted explicit bias remove them without undoing specified first-order flavor protections? No dimensionful wall tension, transition temperature, gravitational-wave spectrum, or cosmologically viable repository benchmark is obtained in this release.

The implementation is `research/flavor_cosmology/cp_wall_tools.mjs`; its dependency-free test is `research/flavor_cosmology/test_cp_wall_tools.mjs`. Its examples and validation record are `receipts/flavor_cosmology/conditional_wall_tools.json`. Run `node research/flavor_cosmology/replay.mjs` to reproduce that receipt; replay retains the historical validation record and regenerates the scientific examples. Run `node research/flavor_cosmology/test_cp_wall_tools.mjs` from the repository root. These additions are isolated from the active scalar-loop and mechanism work. The exact mathematical tools were executed in the available V8 runtime; 104 assertions passed; the standalone replay reproduces the receipt byte for byte in V8. The workspace execution server failed to initialize, so neither the delivered Node command, Python, Lean, nor the full repository suite ran in this session. This difference is recorded rather than attributed to another session's tests.

The coordination snapshot is pinned to `a5d0f04aa6c8f666dfd9428747abbbdf3811fe50`, published at 2026-10-06 19:04 UTC. The arithmetic stream there adds elliptic division-fibre theorems, coordinate normalization, halving transport and rational-root completeness. Its predecessor `ff5d04ab56e417bbc359bfe2c11264ac566cc4d1` supplies exact elliptic witnesses and rational division fibres. The published flavor stream is `8f18f0a037d7936f7ffee9aca1a2a2959c19190e`, published at 18:36 UTC: the protected soft-mediator completion and scalar-loop closure. The user reports that a separate flavor session now owns the full source-quartic exact proof, constrained mechanisms, coupled matching and running, leading quantum and higher-operator responses, extended vacuum tests, physical parameters, and its publication. All of those are excluded from this branch's file edits. The monitoring scope is published GitHub changes and user-supplied session updates; uncommitted work and private session execution are not observable.

The existing 48-scalar completion has a positive full local Hessian, complete finite quark masses and nonzero physical weak CP in its chosen benchmark, with stated first-order mass-phase protection. Its 251-coefficient operator census and fixed-mass response do not establish a golden depth or a predicted 66-degree phase. The earlier exact field-content CP intertwiner is an anti-linear involution on the representations that pass its consistency checks, not a class-inverting operation valid for arbitrary added representations. Those are upstream conclusions in

docs/VALENTINER_SOFT_MEDIATION_AND_LOOP_CLOSURE.md and
docs/VALENTINER_CP_AND_GLOBAL_MODES.md, consumed at the pinned commit. Their current numerical vacuum is not promoted here to a classified global minimum.

Let C denote a consistent CP operation on the full theory and let v be a homogeneous stationary field configuration. If the scalar potential and kinetic metric obey $V(C\phi)=V(\phi)$ and the corresponding isometry relation, Cv is stationary with the same energy and the same canonically measured fluctuation spectrum. An internal-symmetry-invariant CP-odd observable O obeys $O(g\phi)=O(\phi)$ and $O(C\phi)=-O(\phi)$. If $O(v)$ is nonzero, v and Cv cannot be in the same orbit of the internal unitary group: assuming $Cv=gv$ gives $O(v)=-O(v)$. For a quark-matched example, a nonzero Jarlskog invariant with nondegenerate spectra is such a diagnostic. It proves orbit distinction, not a count of all global minima. Distinct orbits alone do not prove disconnected vacuum components: a continuous degenerate path could join them. Isolated global minima after gauge reduction supply the additional hypothesis. A positive physical Hessian establishes local isolation; the repository's remaining global search must establish global minimality.

Counting all 1080 Valentiner matrices as independent physical vacua would therefore be wrong if an internal gauge redundancy identifies their configurations. Conversely, quotienting by ordinary internal transformations does not justify erasing spontaneous CP walls. CP reverses spatial orientation, and gauged parity/CP is not ordinary internal discrete gauge symmetry. McNamara and Reece, Reflections on Parity Breaking, arXiv:2212.00039, analyze exactly stable walls in the gauged-parity setting. A gravity embedding and its orientation structure are not supplied by this finite flavor representation calculation. This release leaves that embedding explicit rather than treating the phrase gauged CP as a wall-removal mechanism.

There is an exact energy result independent of any numerical minimizer. In a CP-invariant quantum theory with a compatible measure, regulator and renormalization prescription, CP-even corrections preserve equality of conjugate branch energies. Thresholds, perturbative running, higher-loop terms and CP-even higher-dimensional operators cannot alone create a CP-conjugate vacuum bias. They may change the common vacuum energy, wall profile or transition order; they do not select a sign. The thermal version holds for a CP-symmetric state, including zero CP-odd chemical potentials. A baryon asymmetry, imposed CP-odd background or explicit CP-breaking spurion changes this premise and must be included as a physical input. Full quantum CP consistency for the repository's enlarged field content is not established by its restricted elementary intertwiner alone.

QCD also supplies no automatic loophole. At a CP-symmetric bare θ and in a CP-symmetric medium, its free energy obeys $E(\theta)=E(-\theta)$. Conjugate branches with opposite induced strong phases therefore have equal QCD energies. In the illustrative susceptibility expansion $E(\theta)=\chi \theta^2/2$, adding an independent explicit common θ_0 gives $E(\theta_0+\theta_s)-E(\theta_0-\theta_s)=2 \chi \theta_0 \theta_s$. That bias relies on the explicit input θ_0 and vanishes if θ_s is zero. It is not a mechanism that follows from protected Nelson–Barr matching. Asadi, Homiller, Lu and Reece, Chiral Nelson–Barr Models: Quality and Cosmology, arXiv:2212.03882, discuss the interaction between strong-CP quality, inflation and reheating in concrete models.

The constructive wall result begins with two supplied degenerate global minima and a continuously differentiable path $\gamma(t)$, $0 \leq t \leq 1$, in canonically normalized real fields. Write $W(t)=V(\gamma(t))-V_{\min}$ and $K(t)=\text{the Euclidean squared norm of } \gamma'(t)$. Provided W is nonnegative, minimization of the one-dimensional spatial parametrization of that fixed path gives a candidate interface tension equal to the integral of $\sqrt{2 W K}$. The true infimum of the interface tension cannot exceed the value from this path. Cauchy–Schwarz then gives the useful exact squared bound

$$\sigma_{\min}^2 \leq \sigma_{\text{path}}^2 \leq 2 \left(\int_0^1 W(t) dt \right) \left(\int_0^1 K(t) dt \right).$$

One sees the first step by introducing $s=dt/dz > 0$ in the static energy: its density integrated over t is $K s/2 + W/s$, minimized pointwise at $s=\sqrt{2W/K}$. Endpoint singularities or interior zeros are understood by limiting finite-energy profiles. Additional paths can lower the upper bound; no lower bound follows. The argument assumes the declared degenerate global minima. Metastable minima with a deeper third branch do not by themselves define a stable cosmological wall.

For rational polynomial inputs, the code computes both integrals exactly with BigInt fractions. It checks that W and its derivative vanish at both endpoints, divides W by t squared times $(1-t)$ squared, and checks nonnegative Bernstein coefficients of the residual polynomial. Nonnegative Bernstein coefficients certify nonnegativity on the unit interval;

failure is deliberately inconclusive. This is not a complete polynomial positivity algorithm. The code accepts a caller-supplied W , so it does not independently verify that W is the composition of a multivariate repository potential with γ . Nor does it certify the endpoints' global minimality. Those hypotheses are written into each receipt. Algebraic endpoints and a nonconstant field metric require further exact-field or interval adapters.

For the canonical one-field benchmark $V(a)=(a^2-1)^2/4$ and $\gamma(t)=2t-1$, $W=4t^2(1-t)^2$ and $K=4$. Their integrals are $2/15$ and 4 . The exact upper bound is $\sigma^2 \leq 16/15$. The known kink has $\sigma=2\sqrt{2}/3$, hence $\sigma^2=8/9$; this gives an independent strict check on the variational bound. These are dimensionless toy data. No repository tension has been assigned this value. With general positive λ and v , the same canonical quartic has tension $2\sqrt{2}\lambda v^3/3$, illustrating why a dimensionless flavor Hessian cannot fix an astrophysical wall tension without a physical normalization and scale.

An explicit bias has a separate, precisely bounded algebraic question. Let O_i be specified CP-odd operators and c_i their small real coefficients. To first order, the relaxed branch-energy difference is $b^T c$, where $b_i=2 O_i(v)$, up to the chosen sign of which branch is subtracted. The derivative of a relaxed energy is the operator value at its unperturbed stationary point; the first-order vacuum shift does not contribute because the unperturbed potential gradient vanishes. The vacuum shift itself is $-H^{-1}$ times the summed c_i gradient O_i . A protected observable Q has first-order response equal to its direct coefficient response minus gradient $Q^T H^{-1}$ gradient O_i . Physical matching must provide that direct response and the properly gauge-reduced Hessian; the present module does not calculate either.

Suppose rows of A collect responses required to vanish to first order, for example a supplied strong-CP response and specified CKM responses. Choose an explicit positive definite coefficient metric R and a budget $c^T R c \leq \rho^2$. The mathematical problem is to maximize $|b^T c|$ subject to $A c=0$. Discard redundant rows of A by exact elimination. Set $x=R^{-1}b$ and

$$z=R^{-1}b - R^{-1}A^T(A R^{-1}A^T)^{-1}A R^{-1}b.$$

Then $A z=0$, $z^T R z=b^T z \geq 0$, and the exact maximum squared bias is ρ^2 times $b^T z$. For a nonzero result, the maximizing coefficient vector is $\rho z/\sqrt{b^T z}$. This follows by taking the R -orthogonal projection of x onto $\ker A$ and applying Cauchy–Schwarz in the R metric. The module gives the rational projected direction, exact squared support and all constraint residuals. A vanishing support is a first-order impossibility result under the supplied constraints and coefficient budget, not an all-orders theorem that a cosmology is impossible.

The exact criterion has a useful sharp interpretation. Because R is positive definite, no allowed first-order bias exists exactly when b lies in the row span of A . If a CP-odd operator distinguishes the conjugate vacua but its bias functional is fully tied to the responses being protected, forbidding those responses also forbids that bias. If b has a surviving component in the constrained subspace, an algebraic first-order direction exists. Its physical size, radiative stability, higher-order strong phase, naturalness and full thermal dynamics remain upstream model calculations. Treating a cancellation $A c=0$ as a symmetry-derived Wilson-coefficient relation would exceed this result.

The hand-worked example uses $R=\text{diag}(1,2,3)$, $b=(2,2,2)$, and the response constraint $c_1+c_2=0$. Its projected direction is $(0,0,2/3)$, so the unit-budget squared maximum is $4/3$, versus $22/3$ without the constraint. A duplicate constraint $2c_1+2c_2=0$ changes nothing. In a two-operator example with $R=I$ and $b=(2,2)$, the same response constraint removes all bias: the support is exactly zero. Twenty-four additional positive definite rational examples check that the projected direction satisfies every original constraint, that its R norm equals its bias pairing, and that adding constraints cannot enlarge the support. No rows in these examples are claimed to be the actual repository strong-CP derivative.

The thermal adapter is deliberately a transparent toy. For a single real CP-odd field with $V(a,T)=\lambda(a^2-v^2)^2/4+c T^2 a^2/2$, positive λ , c and v^2 give $T_c^2=\lambda v^2/c$. Below T_c the conjugate minima have $a^2=v^2-c T^2/\lambda$ and remain exactly energy-degenerate. The example $\lambda=2$, $v^2=9$, $c=3$ gives $T_c^2=6$. The actual 48-scalar system has independent portals and mixed thermal masses. Its critical temperature, possible symmetry nonrestoration, transition order, tunneling and nonthermal restoration are open. In a thermal restoration scenario, maximum temperature after inflation matters; the final reheating temperature alone can miss an earlier hotter epoch. Preheating and inflationary fluctuations likewise need their own calculation.

For bookkeeping only, `classicalScalingScreen` records an explicitly declared horizon-curvature estimate: $\rho_{\text{wall}} = \text{area} \cdot \sigma \cdot H$ and $H_{\text{ann}} = \Delta V / (\text{annihilation} \cdot \sigma)$, with reduced Planck mass M . Equating wall energy to $3 M^2 H^2$ gives $H_{\text{dom}} = \text{area} \cdot \sigma / (3 M^2)$. In this convention, annihilation before both a supplied deadline H_d and wall domination asks for ΔV greater than $\text{annihilation} \cdot \sigma \cdot \max(H_d, H_{\text{dom}})$. All quantities, area and annihilation factors are caller inputs. This is a phenomenological screen, not a rigorous cosmological bound or a simulation.

Combining that declared screen with the exact coefficient envelope yields a conditional test: its required ΔV^2 must be below $\rho^2 b^T z$ to have any linearized coefficient direction within the chosen budget. A certified tension upper bound can supply a conservative sufficient threshold within this screen because both threshold terms increase with σ . An upper bound that fails the test cannot establish impossibility: the true tension might be smaller. Likewise, a support zero under homogeneous first-order response constraints does not exclude quadratic-order bias, inflationary removal or a different field content. Classical estimates are not promoted to universal dynamics. Babichev and collaborators, [arXiv:2504.07902](https://arxiv.org/abs/2504.07902), find different bias dependence in their simulated real-scalar model, reinforcing the need to calibrate a physical network before claiming a gravitational-wave prediction.

The bounded handoff to the active mechanism session is now concrete. It can supply a globally classified pair, a dimensionful canonical metric and potential, physical CP consistency for the complete content, a set of allowed CP-odd operators if explicit breaking is admitted, and their direct plus relaxed strong-CP/CKM response rows. The separate cosmology layer can then consume these inputs to compute path tension bounds, constrained bias support and thermal-history conditions without editing its loop algebra or redoing its matching. If exact CP must remain fundamental, the explicit-bias route is unavailable and the remaining cosmological questions are inflationary removal, restoration avoidance and the actual spacetime realization of CP. These are distinct from choosing a locally stable flavor frame.

The contribution is the executable rational interface between supplied flavor protection constraints and cosmological bias, together with a conservative polynomial-path tension bound and a recorded concurrency boundary. CP-domain-wall existence, their standard cosmological problem, variational wall mechanics and constrained quadratic optimization are established mathematics and physics. No claim of a new fundamental wall mechanism, a new CKM prediction or publication priority is made. References are McNamara and Reece, <https://arxiv.org/abs/2212.00039>; Asadi, Homiller, Lu and Reece, <https://arxiv.org/abs/2212.03882>; Saikawa, <https://arxiv.org/abs/1703.02576>; and Babichev et al., <https://arxiv.org/abs/2504.07902>.